

Programming Fundamentals—Coursework

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BSSE 2A / SE-2A **Subject:** Programming Fundamentals (PF) **Instructor:** Umar Khalil

This repository contains a collection of C++ assignments, a mini project, and MCQ practice for the Programming Fundamentals course.



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Assignment 1 – Making Decisions (Tasks 1–20)

Practice problems covering `if`/`else`, `switch`-style chains, logical operators, and nested conditions.

Task 1 – Simple Condition

```
cpp

#include <iostream>
using namespace std;
int main() {
    int x, y;
    cout << "enter value of y: ";
    cin >> y;
    if (y == 10)
        x = 0;
    else
        x = 1;
    cout << "x = " << x;
    return 0;
}
```

Task 2 – Temperature Range Validator

```
cpp
```

```
#include <iostream>
using namespace std;
int main() {
    int temperature;
    cout << "enter temprature: ";
    cin >> temperature;
    if (temperature >= -50 && temperature <= 150)
        cout << "The number is valid";
    else
        cout << "invalid number";
    return 0;
}
```

Task3– Larger/ Smaller of Two Numbers

```
cpp

#include <iostream>
using namespace std;
int main() {
    int a, b;
    cout << "enter the value of a and b: ";
    cin >> a >> b;
    if (a > b)
        cout << "Larger: " << a << endl << "Smaller: " << b;
    else
        cout << "Larger: " << b << endl << "Smaller: " << a;
    return 0;
}
```

Task4– Floating Point Comparison

```
cpp

#include <iostream>
using namespace std;
int main() {
    double a = 10, b = 10;
    a = a + 0.2;
    if (a > b)
        cout << "a is larger";
    else
        cout << "b is larger";
    return 0;
}
```

Task 5 – Safe Division

```
cpp

#include <iostream>
using namespace std;
int main() {
    int num1, num2;
    cout << "enter numbers: ";
    cin >> num1 >> num2;
    if (num2 != 0)
        cout << "Result: " << (float) num1 / num2;
    else
        cout << "Division by zero is not possible.";
    return 0;
}
```

Task 6 – Logical Operators Practice

```
cpp

#include <iostream>
using namespace std;
int main() {
    int a = 2, b = 4, c = 6;
    if (a == 4 || b > 2) cout << "A) True" << endl; else cout << "A) False" <<
    if (6 <= c && a != 3) cout << "B) True" << endl; else cout << "B) False" <<
    if (1 != b && c != 3) cout << "C) True" << endl; else cout << "C) False" <<
    if (a >= -1 || a <= b) cout << "D) True" << endl; else cout << "D) False" <<
    if (!(a > 2)) cout << "E) True" << endl; else cout << "E) False" << endl;
    return 0;
}
```

Task 7 – Race Placement (First Runner)

```
cpp
```

```

#include <iostream>
using namespace std;
int main() {
    int r1, r2, r3;
    cout << "enter r1, r2, r3: ";
    cin >> r1 >> r2 >> r3;
    if (r1 < r2 && r1 < r3)
        cout << "1st: Runner1" << endl;
    else if (r2 < r1 && r2 < r3)
        cout << "1st: Runner2" << endl;
    else
        cout << "1st: Runner3" << endl;
    return 0;
}

```

Task8 – Shape Area Menu

```

cpp

#include <iostream>
using namespace std;
int main() {
    int choice;
    cout << "1. Circle\n2. Rectangle\n3. Triangle\n";
    cin >> choice;
    if (choice == 1) {
        double r;
        cout << "you chose circle" << endl;
        cout << "enter value of r: ";
        cin >> r;
        cout << 3.14159 * r * r;
    } else if (choice == 2) {
        double l, w;
        cout << "you chose Rectangle" << endl;
        cout << "enter length and width: ";
        cin >> l >> w;
        cout << l * w;
    } else if (choice == 3) {
        double b, h;
        cout << "you chose Triangle" << endl;
        cout << "enter base and height: ";
        cin >> b >> h;
        cout << 0.5 * b * h;
    }
    return 0;
}

```

Task 9 – Print Alphabet with ASCII Codes

```
cpp

#include <iostream>
using namespace std;
int main() {
    for (int ch = 'A'; ch <= 'Z'; ch++) {
        cout << (char)ch << " = " << ch << endl;
    }
    return 0;
}
```

Task 10 – Weight Classifier

```
cpp

#include <iostream>
using namespace std;
int main() {
    double mass, weight;
    cout << "Enter Mass: ";
    cin >> mass;
    weight = mass * 9.8;
    cout << "Weight: " << weight << endl;
    if (weight > 1000)
        cout << "Too heavy";
    else if (weight < 10)
        cout << "Too light";
    return 0;
}
```

Task 11 – Sales Bonus Calculator

```
cpp

#include <iostream>
using namespace std;
int main() {
    double sales;
    cout << "Enter Sales: ";
    cin >> sales;
    if (sales > 50000)
        sales = sales + (0.25 * sales) + 250;
    cout << sales;
    return 0;
}
```

Task 12 – Loan Qualification Check

cpp

```
#include <iostream>
using namespace std;
int main() {
    double income;
    int years;
    cout << "enter income and years: ";
    cin >> income >> years;
    if (income >= 35000 && years > 5)
        cout << "Qualified";
    else
        cout << "Not Qualified";
    return 0;
}
```

Task 13 – TV Model Features

cpp

```
#include <iostream>
using namespace std;
int main() {
    int model;
    cout << "enter Model number: ";
    cin >> model;
    if (model == 300)
        cout << "Picture-in-picture, Stereo sound, Remote control";
    else if (model == 200)
        cout << "Stereo sound, Remote control";
    else if (model == 100)
        cout << "Remote control only";
    else
        cout << "enter valid number";
    return 0;
}
```

Task 14 – Magic Date Checker

cpp

```
#include <iostream>
using namespace std;
int main() {
    int m, d, y;
    cout << "enter month, day, and the last 2 digit of year ";
    cin >> m >> d >> y;
    if (m * d == y)
        cout << "MAGIC";
    else
        cout << "NOT MAGIC";
    return 0;
}
```

Task 15 – Commission Rate

```
cpp

#include <iostream>
using namespace std;
int main() {
    double sales, commission;
    cout << "enter sales: ";
    cin >> sales;
    if (sales <= 10000)
        commission = 0.10;
    else if (sales <= 15000)
        commission = 0.15;
    else
        commission = 0.20;
    cout << commission;
    return 0;
}
```

Task 16 – Checking Account Fee Calculator

```
cpp
```

```

#include <iostream>
using namespace std;
int main() {
    double balance, fee = 10;
    int checks;
    cout << "enter balane and checks: ";
    cin >> balance >> checks;
    if (checks < 20)
        fee += checks * 0.10;
    else if (checks <= 39)
        fee += checks * 0.08;
    else if (checks <= 59)
        fee += checks * 0.06;
    else
        fee += checks * 0.04;
    if (balance < 400)
        fee += 15;
    cout << "Fees: " << fee;
    return 0;
}

```

Task 17 – Subscription Price Menu

```

cpp

#include <iostream>
using namespace std;
int main() {
    int choice;
    cout << "Enter choice: ";
    cin >> choice;
    if (choice == 1)
        cout << "249.00";
    else if (choice == 2)
        cout << "199.00";
    else if (choice == 3)
        cout << "49.00";
    return 0;
}

```

Task 18 – Monthly Plan Cost

```

cpp

```



```

#include <iostream>
using namespace std;
int main() {
    int choice, months;
    cout << "enter choice: ";
    cin >> choice;
    if (choice == 4)
        cout << "enter month: ";
    cin >> months;
    if (choice == 1)
        cout << months * 40;
    else if (choice == 2)
        cout << months * 20;
    else if (choice == 3)
        cout << months * 30;
    return 0;
}

```

Task 19 – Data Package Cost Calculator

```

cpp

#include <iostream>
using namespace std;
int main() {
    int pkg;
    double gb, total;
    cout << "enter pkg and gb: ";
    cin >> pkg >> gb;
    if (pkg == 1) {
        total = 39.99;
        if (gb > 4)
            total += (gb - 4) * 10;
    } else if (pkg == 2) {
        total = 59.99;
        if (gb > 8)
            total += (gb - 8) * 5;
    } else if (pkg == 3) {
        total = 69.99;
    }
    cout << total;
    return 0;
}

```

Task 20 – Seconds to Days/Hours/Minutes

```
cpp

#include <iostream>
using namespace std;
int main() {
    int sec;
    cout << "enter seconds: ";
    cin >> sec;
    if (sec >= 86400)
        cout << "Days: " << sec / 86400 << endl;
    if (sec >= 3600)
        cout << "Hours: " << sec / 3600 << endl;
    if (sec >= 60)
        cout << "Minutes: " << sec / 60 << endl;
    return 0;
}
```

Assignment 3 – Part 1: Arrays & Functions

Instructor: Umar Khalil

Problem 1 – Average of an Array

```
cpp
```

```

#include <iostream>
using namespace std;
float calculateAverage(int a[], int size) {
    int sum = 0;
    for (int i = 0; i < size; i++) {
        sum += a[i];
    }
    return (float)sum / size;
}
int main() {
    int n;
    cout << "enter array size: ";
    cin >> n;
    int a[n];
    for (int i = 0; i < n; i++) {
        cout << "enter Array values: ";
        cin >> a[n];
    }
    cout << "the average score of the student is: " << calculateAverage(a, n) <
    return 0;
}

```

Note: In the input loop, `cin >> a[n]` should be `cin >> a[i]` — reading into `a[n]` is out of bounds and never fills the array correctly.

Problem 2 – Largest and Smallest Values in an Array

cpp

```

#include <iostream>
using namespace std;
int calculateLowest(int arr[], int size) {
    int lowest = arr[0];
    for (int i = 1; i < size; i++) {
        if (arr[i] < lowest) {
            lowest = arr[i];
        }
    }
    return lowest;
}
int calculateHighest(int arr[], int size) {
    int highest = arr[0];
    for (int i = 1; i < size; i++) {
        if (arr[i] > highest) {
            highest = arr[i];
        }
    }
    return highest;
}
int main() {
    int n;
    cin >> n;
    int arr[n];
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    cout << "Lowest Score: " << calculateLowest(arr, n) << endl;
    cout << "Highest Score: " << calculateHighest(arr, n);
    return 0;
}

```

Problem3 – Custom Power Function

cpp

```

#include <iostream>
using namespace std;
int calculatePower(int base, int exponent) {
    int result = 1;
    for (int i = 1; i <= exponent; i++) {
        result = result * base;
    }
    return result;
}
int main() {
    int base, exponent;
    cout << "enter value of base: ";
    cin >> base;
    cout << "enter value of exponent: ";
    cin >> exponent;
    cout << base << " raised to the power " << exponent << " is: "
        << calculatePower(base, exponent) << endl;
    return 0;
}

```

Problem4– Voltage Calculator

```

cpp

#include <iostream>
using namespace std;
double calculateVoltage(double current, double resistance) {
    return current * resistance;
}
int main() {
    double current, resistance;
    cout << "enter current: ";
    cin >> current;
    cout << "enter resistance: ";
    cin >> resistance;
    cout << "Total Voltage: " << calculateVoltage(current, resistance) << " V";
    return 0;
}

```

Problem5– Cricket Team Initial Finder

```

cpp

```

```

#include <iostream>
using namespace std;
bool search(char a[], int size, char target) {
    for (int i = 0; i < size; i++) {
        if (a[i] == target) {
            return true;
        }
    }
    return false;
}
int main() {
    char team[11] = {'r', 'b', 's', 'h', 'i', 'z', 'k', 'm', 'q', 'w', 't'};
    char target;
    cout << "enter target Character: ";
    cin >> target;
    if (search(team, 11, target)) {
        cout << "player is in the team.";
    } else
        cout << "player not found.";
    return 0;
}

```

Problem6– Word Length Calculator

```

cpp

#include <iostream>
using namespace std;
int findLength(char word[]) {
    int count = 0;
    while (word[count] != '\0') {
        count++;
    }
    return count;
}
int main() {
    char word[100];
    cout << "enter word: ";
    cin >> word;
    cout << "length of the word is: " << findLength(word) << endl;
    return 0;
}

```

Assignment3– Part 2: Arrays, Sorting & Strings

Instructor: Umar Khalil

Problem 1 – 2nd Highest and 2nd Lowest Values in an Array

cpp

```
#include <iostream>
using namespace std;
int calculate2Low(int a[], int size) {
    for (int i = 0; i < size - 1; i++) {
        for (int j = i + 1; j < size; j++) {
            if (a[i] > a[j]) {
                int num = a[i];
                a[i] = a[j];
                a[j] = num;
            }
        }
    }
    return a[1];
}
int calculate2High(int a[], int size) {
    return a[size - 2];
}
int main() {
    int n;
    cout << "enter number of student: ";
    cin >> n;
    int a[n];
    cout << "enter score: ";
    for (int i = 0; i < n; i++) {
        cin >> a[i];
    }
    cout << "2nd lowest " << calculate2Low(a, n) << endl;
    cout << "2nd highest " << calculate2High(a, n);
}
```

Problem 2 – Sum of an Array

cpp

```

#include <iostream>
using namespace std;
double calculateSum(int a[], int size) {
    double sum = 0;
    for (int i = 0; i < size; i++) {
        sum += a[i];
    }
    return sum;
}
int main() {
    int n;
    cout << "enter the size: ";
    cin >> n;
    int a[n];
    for (int i = 0; i < n; i++) {
        cin >> a[i];
    }
    cout << "the sum " << calculateSum(a, n) << endl;
    return 0;
}

```

Problem3– Temperature Converter (Celsius→ Fahrenheit)

cpp

```

#include <iostream>
using namespace std;
double celsiusToFahrenheit(double celsius) {
    return (celsius * 9 / 5) + 32;
}
int main() {
    double celsius;
    cout << "enter temperature: ";
    cin >> celsius;
    cout << "temperatue of celsius to farenhight is : ";
    cout << celsiusToFahrenheit(celsius);
    return 0;
}

```

Problem4– Leap Year Checker

cpp


```

#include <iostream>
using namespace std;
bool isLeapYear(int year) {
    if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) {
        return true;
    }
    return false;
}
int main() {
    int n;
    cout << "enter year: ";
    cin >> n;
    cout << isLeapYear(n) << endl;
    cout << isLeapYear(2024);
    return 0;
}

```

Problem5– Vowel Counter

```

cpp

#include <iostream>
using namespace std;
int vowels(char word[]) {
    int count = 0;
    for (int i = 0; word[i] != '\0'; i++) {
        if (word[i] == 'a' || word[i] == 'e' || word[i] == 'i' || word[i] == 'o' || word[i] == 'u') {
            count++;
        }
    }
    return count;
}
int main() {
    char word[100];
    cout << "enter word: ";
    cin >> word;
    cout << "Total vowels: " << vowels(word);
    return 0;
}

```

Problem6– Alphabetical Sorter (Bubble Sort on Chars)

```

cpp

```

```
#include <iostream>
using namespace std;
int main() {
    char arr[5];
    for (int i = 0; i < 5; i++)
        cin >> arr[i];
    for (int i = 0; i < 4; i++) {
        for (int j = 0; j < 4 - i; j++) {
            if (arr[j] > arr[j + 1]) {
                char temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
    cout << "Sorted Letters: ";
    for (int i = 0; i < 5; i++)
        cout << arr[i] << " ";
    return 0;
}
```

Assignment 3 – Part 3: Arrays & 2D Arrays

Roll No: SU92-BSSEM-F25-003

Problem 1 – Count Even and Odd Numbers in an Array

cpp

```

#include <iostream>
using namespace std;
int countEvenNumbers(int arr[], int size) {
    int count = 0;
    for (int i = 0; i < size; i++) {
        if (arr[i] % 2 == 0) {
            count++;
        }
    }
    return count;
}
int countOddNumbers(int arr[], int size) {
    int count = 0;
    for (int i = 0; i < size; i++) {
        if (arr[i] % 2 != 0) {
            count++;
        }
    }
    return count;
}
int main() {
    int N;
    cout << "Enter Size of Array: ";
    cin >> N;
    int numbers[N];
    for (int i = 0; i < N; i++) {
        cout << "Enter value in index " << i << ": ";
        cin >> numbers[i];
    }
    int even = countEvenNumbers(numbers, N);
    int odd = countOddNumbers(numbers, N);
    cout << "Number of Even Numbers in Array: " << even << endl;
    cout << "Number of Odd Numbers in Array: " << odd << endl;
    return 0;
}

```

Problem 2 – Display a 2D Array

cpp

```

#include <iostream>
using namespace std;
int main() {
    int R, C;
    cout << "Enter Number of Rows: ";
    cin >> R;
    cout << "Enter Number of Columns: ";
    cin >> C;
    int arr[R][C];
    cout << "Enter Roll Numbers:" << endl;
    for (int i = 0; i < R; i++) {
        for (int j = 0; j < C; j++) {
            cin >> arr[i][j];
        }
    }
    cout << "\nSeating Chart:" << endl;
    for (int i = 0; i < R; i++) {
        for (int j = 0; j < C; j++) {
            cout << arr[i][j] << " ";
        }
        cout << endl;
    }
    return 0;
}

```

Problem3– Find GCD

```

cpp
#include <iostream>
using namespace std;
int findGCD(int a, int b) {
    while (b != 0) {
        int temp = b;
        b = a % b;
        a = temp;
    }
    return a;
}
int main() {
    cout << findGCD(48, 18);
    return 0;
}

```

Problem4 – Batting Strike Rate Calculator

```
cpp

#include <iostream>
using namespace std;
double calculateStrikeRate(int runs, int balls) {
    return (double(runs) / balls) * 100;
}
int main() {
    int runs, balls;
    cout << "Enter Runs and Balls: ";
    cin >> runs >> balls;
    double strikeRate = calculateStrikeRate(runs, balls);
    cout << "Strike Rate: " << strikeRate << endl;
    return 0;
}
```

Problem5 – Get String Length (Custom)

```
cpp

#include <iostream>
using namespace std;
int getLength(char arr[]) {
    int length = 0;
    for (int i = 0; arr[i] != '\0'; i++) {
        length++;
    }
    return length;
}
int main() {
    char name[] = " Student";
    cout << getLength( name) ;
    return 0;
}
```

Problem6 – Copy a Word (CharArray Copy)

```
cpp
```

```

#include <iostream>
using namespace std;
void copyword(char src[], char dest[], int length) {
    for (int i = 0; i < length; i++) {
        dest[i] = src[i];
    }
    dest[length] = '\\0';
}
int main() {
    char original[] = "Code";
    char backup[10];
    copyword(original, backup, 4);
    cout << backup;
    return 0;
}

```

Assignment 3 – 2D Arrays, Overloading & C-Strings

Additional set of 6 programs covering 2D arrays, function overloading, and C-style string manipulation.

Program 1 – Sum of All Elements in a 2D Array

```

cpp

#include <iostream>
using namespace std;
int main() {
    int r, c;
    cout << "Enter rows and columns: ";
    cin >> r >> c;
    int arr[20][20];
    int sum = 0;
    cout << "Enter elements: \\n";
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) {
            cin >> arr[i][j];
            sum = sum + arr[i][j];
        }
    }
    cout << "Sum of all elements = " << sum;
    return 0;
}

```

Program 2 – Subtract All Elements from the First

```

cpp

```

```

#include <iostream>
using namespace std;
int main() {
    int r, c;
    cout << "Enter rows and columns: ";
    cin >> r >> c;
    int arr[20][20];
    cout << "Enter elements: \n";
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) {
            cin >> arr[i][j];
        }
    }
    int result = arr[0][0];
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) {
            if (i == 0 && j == 0)
                continue;
            result = result + arr[i][j];
        }
    }
    cout << "Final Result = " << result;
    return 0;
}

```

Program 3 – Area Calculator (Function Overloading)

cpp

```

#include <iostream>
using namespace std;
double calculateArea(double side) {
    return side * side;
}
double calculateArea(double length, double width) {
    return length * width;
}
int main() {
    double side;
    double length, width;
    cout << "Enter side of square: ";
    cin >> side;
    cout << "Enter length and width of rectangle: ";
    cin >> length >> width;
    cout << "Area of Square: " << calculateArea(side) << endl;
    cout << "Area of Rectangle: " << calculateArea(length, width);
    return 0;
}

```

Program 4 – Bowling Average Calculator

```

cpp

#include <iostream>
using namespace std;
double getBowlingAverage(int runsConceded, int wicketsTaken) {
    if (wicketsTaken == 0) {
        return 0.0;
    }
    return (double) runsConceded / wicketsTaken;
}
int main() {
    int runs, wickets;
    cout << "Enter runs and wickets: ";
    cin >> runs >> wickets;
    cout << "Bowling Average: " << getBowlingAverage(runs, wickets);
    return 0;
}

```

Program 5 – Print a Word Vertically

```

cpp

```



```

#include <iostream>
using namespace std;
void printVertical(char word[], int size) {
    for (int i = 0; i < size; i++) {
        cout << word[i] << endl;
    }
}
int main() {
    char word[50];
    cout << "Enter a word: ";
    cin >> word;
    int size = 0;
    while (word[size] != '\0') {
        size++;
    }
    printVertical(word, size);
    return 0;
}

```

Program 6 – Reverse a Word In-Place

```

cpp

#include <iostream>
using namespace std;
void reverseword(char word[], int size) {
    for (int i = 0; i < size / 2; i++) {
        char temp = word[i];
        word[i] = word[size - 1 - i];
        word[size - 1 - i] = temp;
    }
}
int main() {
    char word[50];
    cout << "Enter a word: ";
    cin >> word;
    int size = 0;
    while (word[size] != '\0') {
        size++;
    }
    reverseword(word, size);
    cout << "Reversed word: " << word;
    return 0;
}

```

Mini Project: Restaurant Billing System

Introduction

The Restaurant Billing System is a beginner-level C++ program used to manage a customer's food order and calculate the final restaurant bill. The program takes customer details, shows a food menu, allows the user to place orders, and then calculates service charges, GST, discount, and total payable amount.

Purpose

The main purpose of this program is to make the billing process easy for a restaurant. It reduces manual calculation and gives the final bill according to selected food items and quantity.

Main Features

- Customer registration
- Food menu display
- Order placing
- Service charges calculation
- GST calculation
- Discount calculation
- Final bill display
- Customer details view

Working of Program

First, the program asks for customer name, contact number, order type, and number of persons. After this, a menu is shown to the user. The user selects food items and enters quantity. The program adds the price of all selected items into the food bill.

When the bill is calculated, the program applies:

- **Service charges:** 10% for Dine-in and 5% for Takeaway
- **GST:** 16%
- **Discount:** based on the food bill amount

Finally, the program displays the complete bill.

Code

```
cpp
```

```

#include <iostream>
using namespace std;

string customerName, contactNumber, orderType;
int persons;

string foodName[8] = {"Chicken Burger", "Zinger Burger", "Pizza Small", "Pizza
                    "Chicken Biryani", "BBQ Platter", "Fries", "Cold Drink"};
int foodPrices[8] = {450, 550, 900, 1800, 380, 1200, 250, 120};
float foodBill = 0;

void registerCustomer() {
    cout << "Enter customer name: ";
    cin.ignore();
    getline(cin, customerName);

    cout << "Enter contact Number: ";
    cin >> contactNumber;

    cout << "Enter order type (Dine-in/Takeaway): ";
    cin >> orderType;

    cout << "Enter number of persons: ";
    cin >> persons;
}

void showMenu() {
    cout << "\n===== Food Menu =====> endl;
    for (int i = 0; i < 8; i++)
        cout << i + 1 << ". " << foodName[i] << " Rs. " << foodPrices[i] << end
}

void placeOrder() {
    int item, quantity;
    char again;
    do {
        showMenu();
        cout << "Enter item number: ";
        cin >> item;
        cout << "Enter quantity: ";
        cin >> quantity;

        if (item >= 1 && item <= 8 && quantity > 0)
            foodBill = foodBill + foodPrices[item - 1] * quantity;
        else
            cout << "Invalid item or quantity!" << endl;
    } while (again != 'n');
}

```

```

        cout << "Order more? (y/n): ";
        cin >> again;
    } while (again == 'y' || again == 'Y');
}

float serviceCharges() {
    if (orderType == "Dine-in" || orderType == "dine-in")
        return foodBill * 0.10;
    else
        return foodBill * 0.05;
}

float gst() {
    return foodBill * 0.16;
}

float discount() {
    if (foodBill >= 3000 && foodBill <= 5000) return foodBill * 0.05;
    else if (foodBill > 5000 && foodBill <= 10000) return foodBill * 0.10;
    else if (foodBill > 10000) return foodBill * 0.15;
    else return 0;
}

void showBill() {
    float service = serviceCharges();
    float tax = gst();
    float disc = discount();
    float total = foodBill + service + tax - disc;

    cout << "\n===== RESTAURANT BILL =====" << endl;
    cout << "Customer Name: " << customerName << endl;
    cout << "Order Type: " << orderType << endl;
    cout << "Number of Persons: " << persons << endl;
    cout << "Food Bill: Rs. " << foodBill << endl;
    cout << "Service Charges: Rs. " << service << endl;
    cout << "GST: Rs. " << tax << endl;
    cout << "Discount: Rs. " << disc << endl;

    if (total > 5000)
        cout << "Free Delivery Added" << endl;

    cout << "Total Payable Amount: Rs. " << total << endl;
    cout << "Enjoy Your Meal :)" << endl;
}

void showDetails() {

```

```

        cout << "\nCustomer Name: " << customerName << endl;
        cout << " Contact Number: " << contactNumber << endl;
        cout << "Order Type: " << orderType << endl;
        cout << "Persons: " << persons << endl;
    }

int main() {
    int choice;
    registerCustomer();
    do {
        cout << "\n1. View Food Menu";
        cout << "\n2. Place Order";
        cout << "\n3. Calculate Bill";
        cout << "\n4. View Customer Details";
        cout << "\n5. Exit";
        cout << "\nEnter choice: ";
        cin >> choice;

        if (choice == 1) showMenu();
        else if (choice == 2) placeOrder();
        else if (choice == 3) showBill();
        else if (choice == 4) showDetails();
        else if (choice != 5)
            cout << "Invalid choice!" << endl;
    } while (choice != 5);
    return 0;
}

```

Dry Run

Input

```

Customer Name : Ali
Contact Number : 03001234567
Order Type    : Dine-in
Persons       : 2

Order:
1. Chicken Burger, Quantity = 2
2. Cold Drink,      Quantity = 2

```

Calculation

Chicken Burger price = 450 → 2 burgers = 450 × 2 = 900

Cold Drink price = 120 → 2 drinks = 120 × 2 = 240

Food Bill = 900 + 240 = 1140

Service Charges = 1140 × 10% = 114

GST = 1140 × 16% = 182.4

Discount = 0 (bill is less than 3000)

Total Bill = 1140 + 114 + 182.4 - 0 = 1436.4

Output

```
Customer Name      : Ali
Order Type         : Dine-in
Number of Persons  : 2
Food Bill          : Rs. 1140
Service Charges    : Rs. 114
GST                : Rs. 182.4
Discount           : Rs. 0
Total Payable Amount : Rs. 1436.4
```

MCQs – Chapters 2, 3 & 4

Chapter 2: Introduction to C++

1. Which data type stores whole numbers? A) float B) int C) char D) string **Answer: B**
2. What is the result of assigning 3.88 to an int variable? A) 3.88 B) 4 C) 3 D) Error **Answer: C**
3. Which is a character constant? A) "A" B) 'A' C) A D) "A' **Answer: B**
4. What does `endl` do? A) Ends program B) Starts loop C) Moves to next line D) Deletes output **Answer: C**
5. What is a variable? A) Fixed value B) Memory location with changing value C) Operator D) Function **Answer: B**

Chapter 3: Expressions and Interactivity

1. What is `static_cast` used for? A) Looping B) Type conversion C) Input D) Output **Answer: B**
2. Overflow occurs when: A) Value is too small B) Value is too large C) Value is correct D) Value is zero **Answer: B**
3. Which is a named constant? A) `int x` B) `const int x = 5` C) `x = 5` D) `5` **Answer: B**
4. What does the `+=` operator do? A) Multiply B) Add and assign C) Divide D) Compare **Answer: B**

5. Which function reads a full line (with spaces)? A) `cin >>` B) `getline()` C) `input()` D) `read()` **Answer: B**

Chapter 4: Making Decisions

1. Which logical operator means AND? A) `||` B) `&&` C) `!` D) `==` **Answer: B**
2. Which value represents false? A) 1 B) -1 C) 0 D) 2 **Answer: C**
3. What does `switch` use? A) float B) string C) integer/char D) array **Answer: C**
4. What is the purpose of `break` in `switch`? A) Continue loop B) Exit switch C) Start program D) Compare values **Answer: B**
5. What is an `enum`? A) Loop B) Function C) User-defined data type D) Operator **Answer: C**
-

How to Compile & Run

Each program can be compiled individually with `g++`:

```
baSh
g++ -o program program.cpp
./program
```

Notes

- Some snippets above were transcribed and may contain small syntax slips (e.g., been kept faithful to the original source).
- This repository is intended for personal use in the C++ Fundamentals course.